

Anirudh Jayaprakash

+1 (240) 726-0258 | anirudhjp313@gmail.com | linkedin.com/in/anirudh-jp | github.com/AniJP

Professional Summary

Machine Learning Engineer with hands-on experience building diffusion models, multi-agent AI systems (CrewAI, LangChain), and production ML pipelines. Published researcher (IEEE Xplore, GCAT 2023) with expertise in computer vision, NLP, and GenAI. Proficient in Python, PyTorch, TensorFlow, SQL, and cloud-native deployment using Docker, AWS, and CI/CD.

Technical Skills

Languages: Python, C, C++, Java, SQL, R

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Diffusion Models, U-Net, Swin Transformers, LLMs, Generative AI

LLM & Agentic AI: RAG, Prompt Engineering, LangChain, CrewAI, Vector Embeddings, Function Calling

Data Engineering & Big Data: ETL/ELT Pipelines, Distributed Data Processing, Apache Spark, Databricks, Data Lakes, Data Validation

Cloud & MLOps: AWS, Azure, Docker, Jenkins, CI/CD, MLflow, Git

Web, APIs & Databases: Flask, Gradio, REST APIs, Postman, Selenium, PostgreSQL, MySQL, MongoDB, SQLite, pgvector, ChromaDB

Projects

Multi-LoRA Adapter Serving with vLLM | [GitHub](#)

September 2026

- Fine-tuned 3 task-specialized LoRA adapters (SQL generation, dialogue summarization, NER-to-JSON extraction) on a shared Llama-3.2-3B base using PEFT/TRL, then served all 3 simultaneously from a single vLLM engine via LoRARequest routing, eliminating reloading and duplicate model copies.
- Debugged and resolved 5+ compatibility issues across the PyTorch/vLLM/trl stack (GPU kernel mismatches, KV-cache memory limits, deprecated dataset loading, library API drift), documenting root causes and fixes for reproducibility.
- Demonstrated measurable behavioral gains from task-specific adapters over the generic base model, e.g., the extraction adapter emitted clean, directly parseable JSON where the base model wrapped output in markdown and prose that broke naive parsing.

Diffusion-Based Image Super-Resolution with ResShift | [GitHub](#)

November 2025 – January 2026

- Engineered a diffusion-based super-resolution architecture with U-Net and Swin Transformer blocks for 4x image enhancement.
- Integrated VQGAN latent-space processing to improve generation quality and computational efficiency.

NBA Insights API — Multi-Agent LLM System | [GitHub](#)

October 2025 – December 2025

- Built a multi-agent NBA Q&A system with CrewAI, FastAPI, and Pandas, exposing a REST API with 5 structured data tools; achieved 100% tool-use rate on a golden eval set.
- Added reliability guardrails to the agent pipeline (groundedness gate, TTL caching, retry logic, tool-call budgeting), reaching 1.0 faithfulness and 92% tool-call success at 2.7s p50 latency.

Experience

University of Maryland, College Park

September 2025 – May 2026

Graduate Assistant

- Built Retrieval-Augmented Generation (RAG) pipelines using LangChain and CrewAI with OpenAI API integration to automate literature and data analysis tasks for research projects.
- Built and benchmarked a local LLM inference stack (Ollama, FastAPI) for the research team, evaluating models on time-to-first-token, decode throughput, and latency — identified up to 6x throughput differences across model choices to inform tooling decisions.
- Performed statistical analysis and hypothesis testing (Python: pandas, SciPy/statsmodels; R) on research datasets, building visualizations and dashboards to communicate findings to the research team.

Urban Ventures

October 2023 – March 2024

Web Developer Intern

India

- Developed responsive web interfaces using JavaScript, jQuery, and Bootstrap, increasing user engagement by 30%.
- Integrated REST APIs and optimized mobile rendering, reducing bounce rate by 25%.

lastminute.com

February 2023 – August 2023

Software Developer Intern

India

- Automated QA workflows through modular Selenium and Appian test suites, reducing manual QA effort by 45%.
- Refactored legacy tests into data-driven scripts, improving runtime efficiency by 30%.
- Validated REST APIs and supported Jenkins CI/CD pipelines to accelerate release cycles.

Education

University of Maryland, College Park

August 2024 – May 2026

M.S. in Data Science

College Park, MD

- Relevant Coursework: Advanced Machine Learning, Big Data Infrastructure, Deep Learning, NLP, Neural Networks

CMR Institute of Technology (CMRIT)

2019 – 2023

B.E. in Information Science

Bangalore, India

Publications

A. Jayaprakash et al., “Knowledge Graph Based Medical Chatbot Building,” in Proc. 2023 4th IEEE Global Conference for Advancement in Technology (GCAT), 2023. View on IEEE Xplore